AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 13 · WORKBOOK

State machines for bounded agents

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete bounded-state policy and trace record.

Keep graph version, guards, counters, accepted prefix, first rejection and actual evidence scope together.

Keep one habit: Recall → W1–W3/W5 → independent graph → project state policy → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A bounded agent uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible.

- **State**s are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Logs traces and privacy

> **My remembered idea, example and correction**

Evaluation pipelines

> **My remembered idea, example and correction**

Testing pure components

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main original graph: ready→looking_up; looking_up→review or done; review→done only after authorized review evidence; done terminal. Start ready/counter 0; cap 2 accepted lookup-state entries does not add retry edges. Proposed ready→looking_up→review→looking_up→done first rejects review→looking_up. Accepted prefix has one lookup entry. Valid review prefix can remain incomplete.

  Review→done needs the supplied trusted evidence flag in the extended local audit; flag not real verification. A separately copied graph may explicitly add review→looking_up, allowing a second entry, but a third still exhausts the retained budget; original graph stays unchanged. Separate parcel permits idle→querying→checking→complete; path B querying→complete invalid.

  Adding checking→querying does not authorize that shortcut. Compact original helper checks adjacency only; no actual tool or fetched-data correctness is demonstrated.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main original graph: ready→looking_up; looking_up→review or done; review→done only after authorized review evidence; done terminal. Start ready/counter 0; cap 2 accepted lookup-state entries does not add retry edges. Proposed ready→looking_up→review→looking_up→done first rejects review→looking_up. Accepted prefix has one lookup entry. Valid review prefix can remain incomplete.

  Review→done needs the supplied trusted evidence flag in the extended local audit; flag not real verification. A separately copied graph may explicitly add review→looking_up, allowing a second entry, but a third still exhausts the retained budget; original graph stays unchanged. Separate parcel permits idle→querying→checking→complete; path B querying→complete invalid.

  Adding checking→querying does not authorize that shortcut. Compact original helper checks adjacency only; no actual tool or fetched-data correctness is demonstrated.

- Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main original graph: ready→looking_up; looking_up→review or done; review→done only after authorized review evidence; done terminal. Start ready/counter 0; cap 2 accepted lookup-state entries does not add retry edges. Proposed ready→looking_up→review→looking_up→done first rejects review→looking_up. Accepted prefix has one lookup entry. Valid review prefix can remain incomplete.

  Review→done needs the supplied trusted evidence flag in the extended local audit; flag not real verification. A separately copied graph may explicitly add review→looking_up, allowing a second entry, but a third still exhausts the retained budget; original graph stays unchanged. Separate parcel permits idle→querying→checking→complete; path B querying→complete invalid.

  Adding checking→querying does not authorize that shortcut. Compact original helper checks adjacency only; no actual tool or fetched-data correctness is demonstrated.

- A report omits this qualification: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main original graph: ready→looking_up; looking_up→review or done; review→done only after authorized review evidence; done terminal. Start ready/counter 0; cap 2 accepted lookup-state entries does not add retry edges. Proposed ready→looking_up→review→looking_up→done first rejects review→looking_up. Accepted prefix has one lookup entry. Valid review prefix can remain incomplete.

  Review→done needs the supplied trusted evidence flag in the extended local audit; flag not real verification. A separately copied graph may explicitly add review→looking_up, allowing a second entry, but a third still exhausts the retained budget; original graph stays unchanged. Separate parcel permits idle→querying→checking→complete; path B querying→complete invalid.

  Adding checking→querying does not authorize that shortcut. Compact original helper checks adjacency only; no actual tool or fetched-data correctness is demonstrated.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Only these **transition**s are allowed: ready→looking_up→review→done. Done is terminal. Evaluate ready→review→done and ready→looking_up→review→done.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **state** machines for bounded agents **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **state** machines for bounded agents.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **State** machines for bounded agents. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **State** machines for bounded agents. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid **transition**.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal.

- A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

- Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal.
- Which labelled step matches this detail? Querying→complete skips checking and is invalid.
- 1. Follow A
- 2. Read arrows
- 3. Inspect B

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect B Querying→complete skips checking and is invalid. A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid.

Changed-case question and optional hint

- Changed-case practice: Add checking→querying as a permitted retry. Which conclusion is supported by the supplied facts?
- 1. That retry becomes valid; querying→complete stays forbidden.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That retry becomes valid; querying→complete stays forbidden. Adding one arrow does not authorise unrelated shortcuts.

Repair and retain the record

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

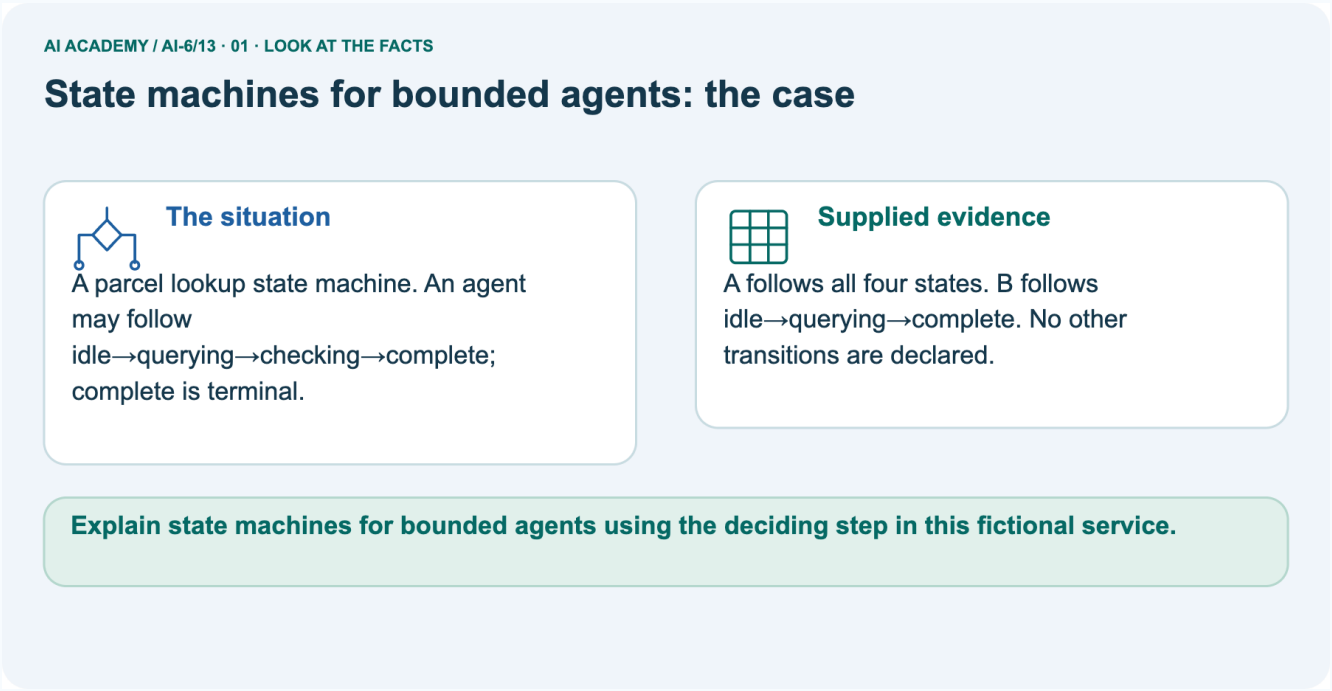
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. State machines for bounded agents: the case



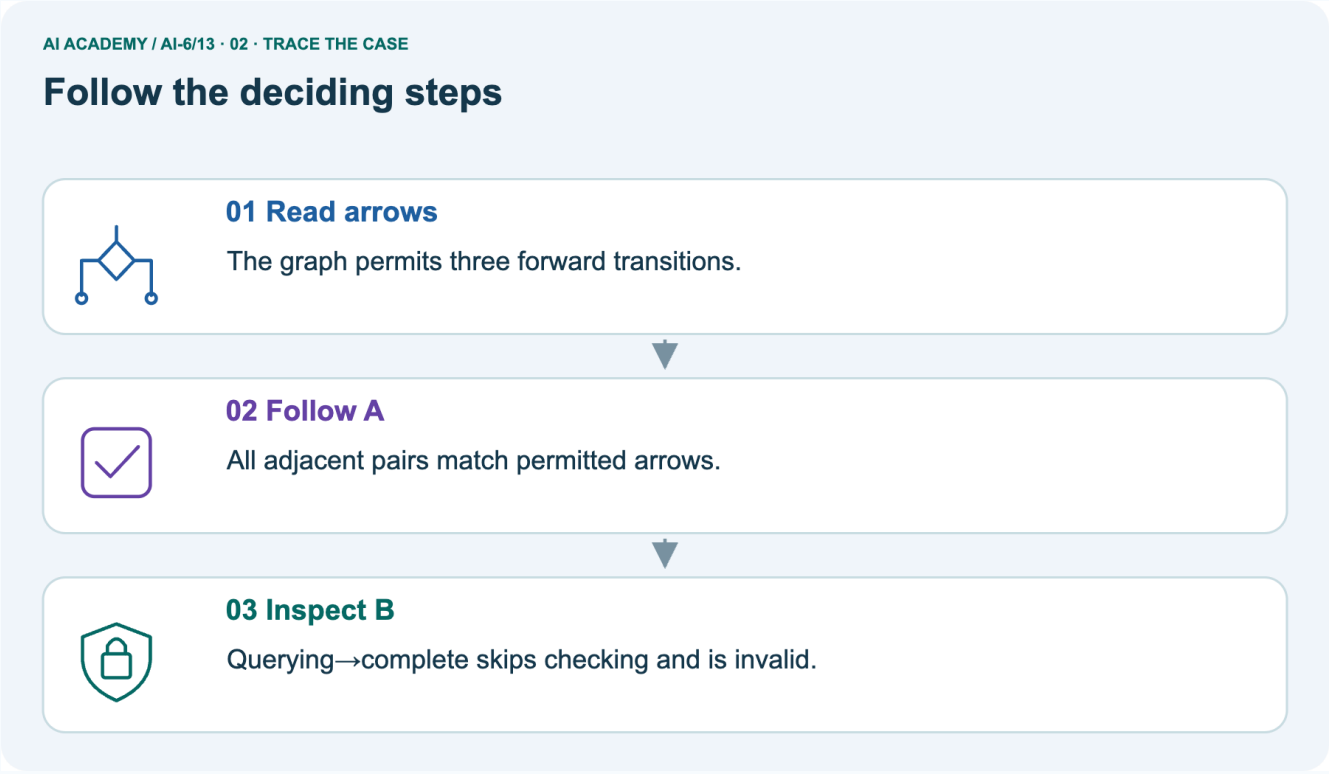
Follow the labelled picture with the matching explanation. Open the vector diagram

A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal. Supplied facts: A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read arrows: The graph permits three forward **transition**s. → Follow A: All adjacent pairs match permitted arrows. → Inspect B: Querying→complete skips checking and is invalid.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. State machines for bounded agents — evidence trace

AI ACADEMY / AI-6/13 · 03 · INSPECT THE LESSON MODEL

State machines for bounded agents — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Trace ready -> looking_up -> review -> looking_up -> done.	States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.	review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

State machines for bounded agents — **evidence** trace.

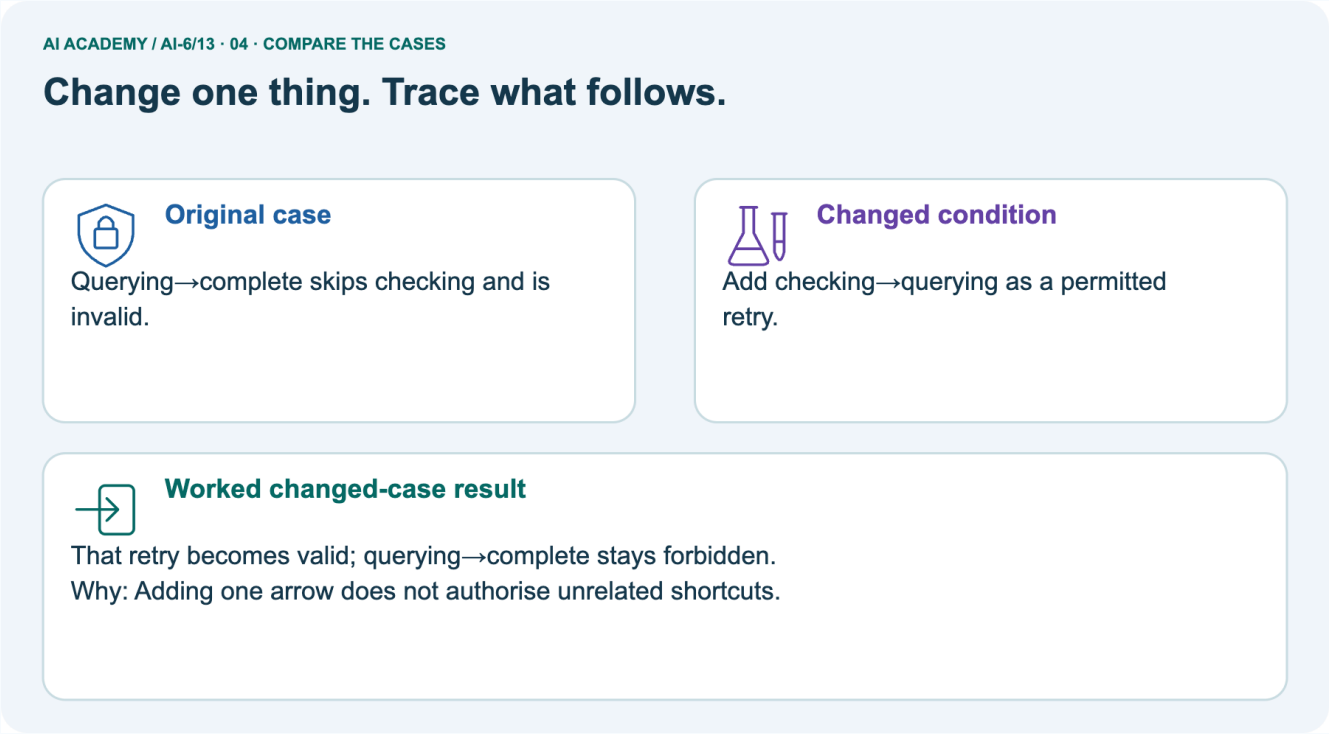
Trace ready -> looking_up -> review -> looking_up -> done. | **State**s are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal.

Maximum two lookup attempts. | review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

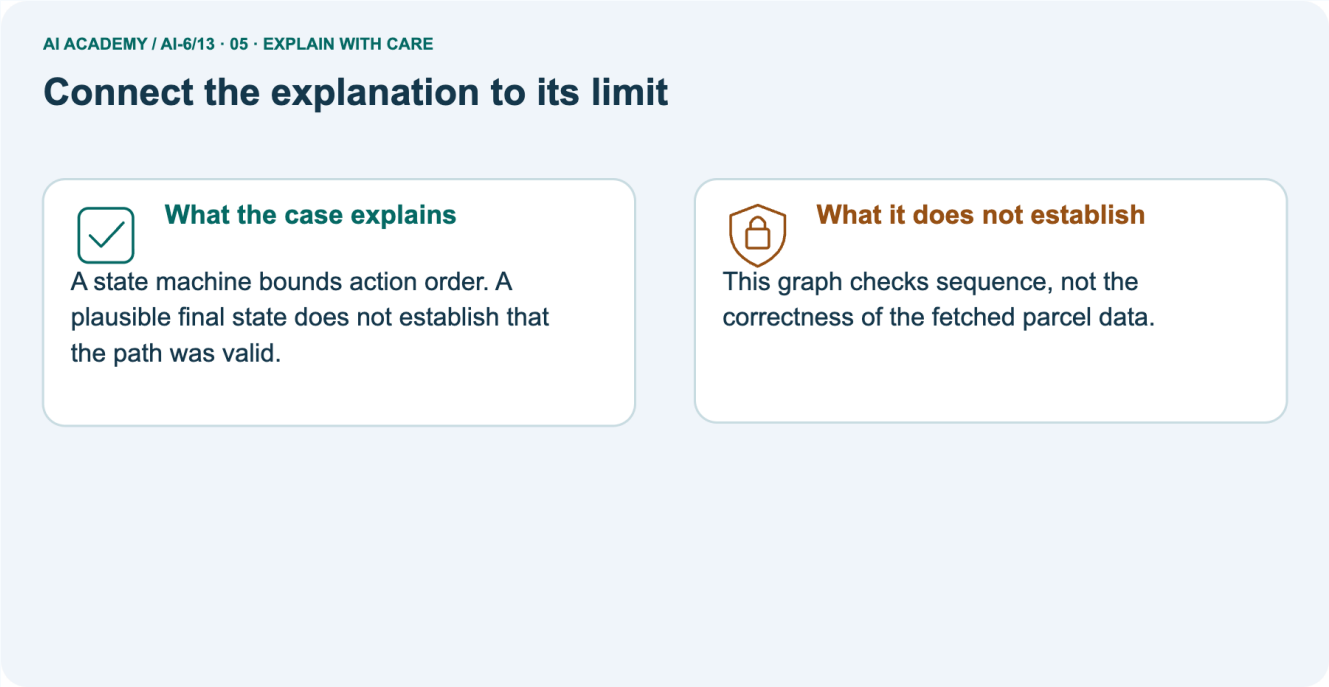
Original case: Querying→complete skips checking and is invalid. Changed condition: Add checking→querying as a permitted retry. Expected result: That retry becomes valid; querying→complete stays forbidden.

Reason: Adding one arrow does not authorise unrelated shortcuts.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



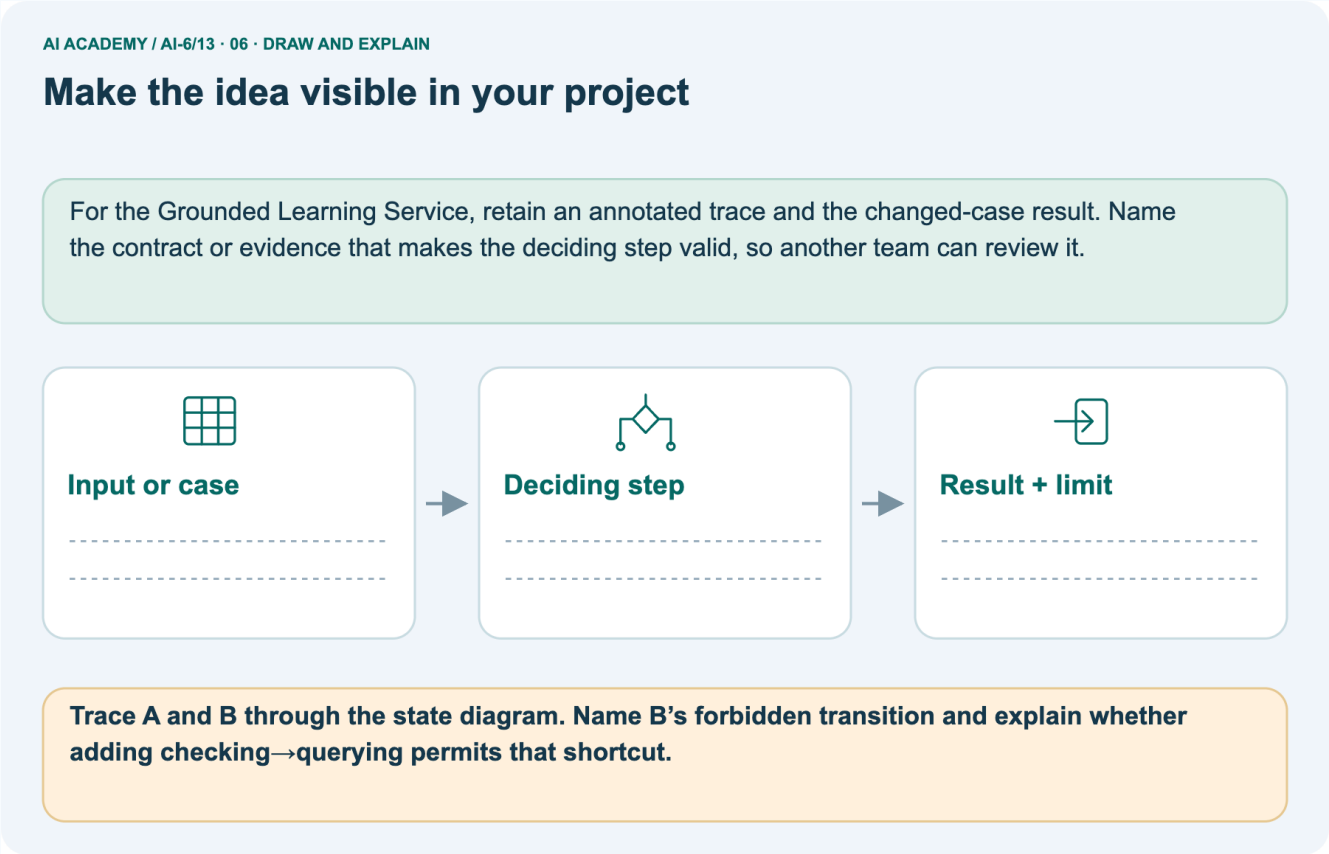
Follow the labelled picture with the matching explanation. Open the vector diagram

A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid. Limit: This graph checks sequence, not the correctness of the fetched parcel data.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. State machines for bounded agents: the case

02. Follow the deciding steps

03. State machines for bounded agents — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.